

Sara Rostami

MSc Candidate, Neuroscience (Machine Learning Specialization)
University of Western Ontario & Vector Institute — Mohsenzadeh Lab
London, Ontario, Canada · +1-416-834-7801 · sararostami.d98@gmail.com
[linkedin.com/in/sara-rostami-7022181b0](https://www.linkedin.com/in/sara-rostami-7022181b0) · github.com/SaraRostami · saraostami.github.io

RESEARCH INTERESTS

Visual perception and image memorability · multimodal neuroimaging (simultaneous EEG–fNIRS) · neuro-inspired and comparative deep network models · model interpretability, uncertainty quantification, and calibration.

EDUCATION

Master of Science in Neuroscience (Machine Learning Specialization) Sep 2025 – Aug 2027
University of Western Ontario, Canada GPA 3.97 / 4.0

Thesis: Neural underpinnings of visual category encoding and image memorability, investigated with simultaneous EEG–fNIRS recordings.
Supervisor: Dr. Yalda Mohsenzadeh.

Master of Science in Artificial Intelligence and Robotics Sep 2021 – 2024
University of Tehran, Iran — ranked 3rd in cohort GPA 19.12 / 20 (≈3.9 / 4.0)

Bachelor of Science in Computer Engineering Sep 2016 – 2021
Babol Noshirvani University of Technology, Iran GPA 17.38 / 20

RESEARCH EXPERIENCE

Graduate Researcher (MSc Thesis) Sep 2025 – Present
Mohsenzadeh Lab, University of Western Ontario & Vector Institute, Canada

- Designed and run a simultaneous EEG/fNIRS acquisition protocol — 32-channel EEG co-registered with 32 fNIRS optodes — recording 20+ participants across 5 image categories.
- Built the study’s end-to-end preprocessing, artifact-rejection, and feature-extraction pipeline, replacing a manual workflow with one reproducible run per subject across all datasets.
- Ran multivariate decoding over 936 trials per participant with permutation-based significance testing, linking neural signals to behavioural measures of visual category perception and image memorability.
- Benchmarking learned representations from vision models (AlexNet, ResNet, and larger pretrained backbones) against the recorded neural data to characterize how memorability is encoded across network depth.

Research Assistant Aug 2022 – Aug 2025
Convergent Technologies Research Center, University of Tehran, Iran

- Collected and preprocessed 128-channel EEG from 15 participants for neural adaptation studies, applying MVPA to identify cognitive correlates of repetition and expectation effects in visual mismatch negativity.
- Applied univariate and multivariate analyses with permutation testing to isolate spatiotemporal neural signatures, informing downstream computational models of visual mismatch negativity.

Research Intern Aug 2022 – Feb 2023
Genzel Lab, Donders Institute, Radboud University, Netherlands

- Compared four classifiers (logistic regression, decision tree, random forest, naive Bayes) on physiological sleep recordings, using random-forest feature importance to identify key predictors of memory consolidation.
- Engineered features from raw physiological time series and produced publication-ready figures on systems consolidation during sleep.

TEACHING EXPERIENCE

Graduate Teaching Assistant — University of Western Ontario, Canada Winter – Fall 2026

- Introduction to Programming in Python (CS1026A) — supported students with assignments and course concepts; graded via Gradescope.
- Object-Oriented Programming with Java — led labs and tutorials for an undergraduate course.

Teaching Assistant — Neuromatch Academy, Computational Neuroscience Summer School 2023, 2025, 2026

- Project Teaching Assistant (2025, 2026) — supported student project pods with dataset handling, analysis workflows, and project development.
- Teaching Assistant (2023) — mentored students through tutorials and projects in the computational neuroscience curriculum.

Graduate Teaching Assistant — University of Tehran, Iran Spring 2023

- Statistical Inference (graduate) — designed and graded assignments and supported students with course concepts.
- Introduction to Cognitive Neuroscience (graduate) — assisted with tutorials, grading, and core concepts.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, MATLAB, Java, HTML/CSS

Machine Learning & Deep Learning: PyTorch, Scikit-learn, Keras, SciPy — MVPA, Bayesian inference, uncertainty quantification & calibration, explainable AI (SHAP, LIME, D-RISE), fine-tuning & domain adaptation

Generative AI & LLM: Hugging Face, LangChain, LangGraph, OpenAI / Gemini APIs — retrieval-augmented generation, agentic workflows, prompt engineering, model evaluation

Data & Development: PostgreSQL, SQL Server, Psycogp2, FastAPI, Git, pandas, NumPy, Selenium, BeautifulSoup

Visualization: Matplotlib, Seaborn, Plotly

PROJECTS

Calibrated Uncertainty in Vision-Language Models: Benchmarked post-hoc calibration methods for a BLIP-2 vision-language model on VQA v2, comparing temperature scaling against a Laplace approximation (Laplace-Torch). Temperature scaling reduced expected calibration error from 0.321 to 0.097, outperforming the Bayesian approach. Probabilistic Generative AI course project. [GitHub](#)

Neutro-Sentry — Hack Western 12: Real-time multimodal stroke-risk triage tool built end-to-end in a 36-hour hackathon: a SwiftUI iOS client streams facial-mesh landmarks and vitals via the Presage SmartSpectra SDK to a Python/FastAPI backend, where a Gemini pipeline grounded in embedded clinical guidelines returns structured risk scores and explanations to a React/TypeScript dashboard. [Devpost](#)

Explainable AI: Model Interpretability with SHAP, D-RISE, and LIME: SHAP (Deep/Kernel) on a WHO life-expectancy network, identifying schooling as the dominant predictor via force plots; D-RISE for model-agnostic explanations in object detection (localization and classification); LIME superpixel saliency on MobileNetV2. [GitHub](#)

Attention Mechanisms & Transformers: BERT Encoder and BEiT: Implemented a BERT encoder from first principles with scaled dot-product and multi-head attention, visualizing learned attention weights on sample sentences; fine-tuned BEiT for vision segmentation, reaching 0.65 IoU. [GitHub](#)

Fraud, Liveness Detection, and OCR with Deep Networks: SMOTE with an autoencoder for fraud detection (87% recall); LeNet-5 for liveness detection (95% validation accuracy); a custom CNN for OCR (99% accuracy, Adam optimizer). [GitHub](#)

GANs for Image Synthesis: Implemented DCGAN, AC-GAN, and WGAN on sign-language MNIST; WGAN-GP with gradient penalty and label smoothing achieved stable convergence while mitigating mode collapse, producing diverse, high-fidelity synthetic samples. [GitHub](#)

Cryptocurrency Price Prediction: Scraped CoinMarketCap data with Selenium; a CNN ensemble with technical indicators reached 92% accuracy on crypto trend prediction. [GitHub](#)

Customer Retention Analysis: Cohort heatmap analysis on e-commerce data showing 48% peak retention; recommended September/October campaigns for a projected 15% uplift. [GitHub](#)

Musical Instrument Classification: Curated a dataset of 9,000+ Persian instrument audio clips (ney, piano, santur, setar, tar, violin); extracted MFCC (20 coefficients) and zero-crossing features with librosa; SVM (RBF kernel) and MLP reached 84% test accuracy. [GitHub](#)

HONORS AND AWARDS

Vector Institute Research Grant \$4,000 grant for graduate students of Vector-affiliated faculty, recognizing excellence in AI research potential.	2025 – 2026
Western University Graduate Funding Package Competitive funding award valued at \$36,950 per year for two years, University of Western Ontario.	2025 – 2027
Top Ranked among Master’s Students Ranked 3rd in the Artificial Intelligence and Robotics program, University of Tehran.	2024
Ranked 98th in National Master’s Entrance Exam Admitted to the University of Tehran as one of 18 students selected from over 14,000 participants.	2021